Dissertation Appendices Draft

Appendix A: Data Collection Workflow, Retrieval Conditions, and Data Cleaning

This appendix documents the collection and preparation of the public player review corpus used in the dissertation. The main dataset was collected from three review platforms: Steam, App Store, and Google Play. These platforms were selected because they provide publicly available player reviews and include standardised evaluative markers that can be converted into sentiment labels. Steam reviews provide a recommended / not recommended judgement, while App Store and Google Play reviews provide star ratings.

A.1 Platforms and Games

Game	Platform	Retrieval condition	Reviews retained
Hakuoki: Kyoto Winds	Steam	App ID `589530`; languages: English, Simplified Chinese, Traditional Chinese, Japanese	1,976
Monster Prom	Steam	App ID `743450`; languages: English, Simplified Chinese, Traditional Chinese, Japanese	12,629
Love and Deepspace	App Store	App Store regional pages for US, CN, TW, and JP	1,999
Love and Deepspace	Google Play	Package name `com.papegames.lysk.en`; language queries and post-collection language filtering	25,463

The final baseline corpus contains 42,067 reviews: 1,976 for Hakuoki: Kyoto Winds, 27,462 for Love and Deepspace, and 12,629 for Monster Prom. The language distribution is English 36,768, Simplified Chinese 4,099, Traditional Chinese 672, and Japanese 528.

A.2 Steam Review Collection

Steam reviews were collected for Hakuoki: Kyoto Winds and Monster Prom. The Steam app IDs were used to retrieve public reviews in English, Simplified Chinese, Traditional Chinese, and Japanese. The collection did not set a fixed upper limit; the retained dataset reflects the reviews that were publicly retrievable at the time of collection.

For Steam reviews, the following personal or account-identifying fields were excluded from the final analytical files: `steamid`, `profile_url`, `avatar`, `persona_name`, `author_num_games_owned`, and `author_num_reviews`. The retained fields included the review text, language, recommendation label, voting metadata, timestamps, and anonymised playtime fields. The Steam recommendation label was converted into the baseline sentiment label.

A.3 App Store Review Collection

App Store reviews were collected for Love and Deepspace across four storefront regions: US, CN, TW, and JP. The collection used the regional app pages and corresponding app IDs. Reviews were retained in English, Simplified Chinese, Traditional Chinese, and Japanese.

Personal author fields were removed before analysis, including `author`, `author_uri`, and `author_label`. The retained fields included title, review text, language, storefront, anonymised review ID, rating, version, and update date. Star ratings were mapped into positive, neutral, and negative sentiment categories.

A.4 Google Play Review Collection

Google Play reviews were collected for Love and Deepspace using the package name `com.papegames.lysk.en`. Because the Google Play language request parameter does not guarantee that every returned review belongs to the requested language, the collected reviews were filtered after collection using heuristic language checks. Japanese was identified through kana characters, English through ASCII-letter ratio, and Simplified / Traditional Chinese through common lexical and character-form differences.

Personal user fields were excluded, including `userName` and `userImage`. Retained fields included title, review text, language, query language, country, anonymised review ID, star score, thumbs-up count, review version, date fields, and developer reply fields where available.

A.5 Cleaning and Preparation

Across all platforms, the cleaning process followed four principles. First, personal identifiers were removed before the data entered the analytical workflow. Second, each review was normalised into a shared structure including game, source, language, record ID, text, rating or recommendation marker, sentiment label, and sentiment score. Third, reviews were grouped by game, platform, and language to support cross-case and cross-language comparison. Fourth, the cleaned corpus was prepared for baseline sentiment analysis, aspect-based sentiment analysis, thematic coding, and exploratory topic modelling.

Appendix B: Thematic Coding Framework

The thematic coding framework was developed to examine how players describe narrative form, romance structure, gameplay structure, emotional response, and agency or constraint. Codes were used as interpretive labels during close reading, not as automatic keyword categories. Each review could receive primary and secondary codes where relevant.

B.1 Coding Procedure

1. Read the whole review once without assigning codes.
2. Identify whether the review mainly discusses narrative form, romance structure, gameplay structure, emotional response, agency, or a combination of these categories.
3. Assign primary and secondary codes in the relevant categories.
4. Write a short researcher memo explaining why the review matters for the dissertation question.
5. Mark whether the review should be quoted, paraphrased, or excluded from dissertation writing.

B.2 Codebook

Category	Code	Meaning
Narrative form	`N1_route_structure`	Routes, route order, branching paths, route split, or character-specific story organisation
Narrative form	`N2_pacing_and_payoff`	Slow or fast pacing, delayed romance, incomplete story payoff, endings, or satisfaction after narrative build-up
Narrative form	`N3_worldbuilding_lore_history`	Historical setting, lore, timeline, worldbuilding, myth, or background exposition
Narrative form	`N4_protagonist_perspective`	Heroine / MC, self-insert, identification, voice, personality, or narrative point of view
Narrative form	`N5_fragmented_storytelling`	Story distributed through cards, events, memories, updates, optional episodes, or scattered fragments
Romance structure	`R1_attachment_to_love_interest`	Attachment, affection, devotion, comfort, desire, or favourite-character preference
Romance structure	`R2_romance_payoff`	Whether romantic scenes, intimacy, route endings, or emotional reward feel satisfying
Romance structure	`R3_romance_disappointment`	Romance described as shallow, delayed, weak, interrupted, or emotionally

		unconvincing
Romance structure	R4_character_chemistry	Chemistry, relationship development, interactions, banter, or emotional dynamic
Romance structure	R5_female_player_positioning	Self-insertion, female desire, agency, gendered fantasy, or discomfort with player positioning
Romance structure	R6_performance_and_affective_continuity	Voice acting, performer continuity, audiovisual intimacy, or performance changes that affect attachment
Gameplay structure	G1_choice_and_branching	Choices, branching, stats, route requirements, or meaningfulness of choices
Gameplay structure	G2_replay_and_completion	Replay, completionism, multiple endings, route order, repeated runs, or unlocking content
Gameplay structure	G3_randomness_and_RNG	Randomness, event pools, luck, unpredictable outcomes, or procedural route access
Gameplay structure	G4_combat_cards_progression	Combat, cards, levelling, story gates, battle difficulty, or progression barriers
Gameplay structure	G5_monetisation_and_resources	Gacha, payment, monetisation, stamina, resources, currency, or free-to-play constraints
Gameplay structure	G6_technical_accessibility	Bugs, crashes, device requirements, localisation problems, readability, or access barriers
Emotional response	E1_attachment_affection	Love, fondness, comfort, longing, protectiveness, crying, or strong attachment
Emotional response	E2_immersion_identification	Immersion, self-insertion, identification, being drawn in, or feeling present in the story

Emotional response	`E3_frustration_irritation`	Frustration, annoyance, anger, fatigue, boredom, or irritation
Emotional response	`E4_disappointment_ambivalence`	Mixed or disappointed response despite interest in characters, story, or premise
Emotional response	`E5_satisfaction_enjoyment`	Enjoyment, satisfaction, fun, appreciation, or recommendation
Emotional response	`E6_confusion_cognitive_labor`	Confusion, effort to understand timelines/lore/systems, theorising, or interpretive labour
Emotional response	`E7_motivation_to_continue_or_replay`	Desire to continue, replay, unlock content, complete routes, or return to the game
Agency / constraint	`A1_agency_enhanced`	Structure described as increasing agency, choice, emotional control, or meaningful pursuit
Agency / constraint	`A2_agency_limited`	Structure described as limiting agency through gates, RNG, monetisation, route locks, or weak choices
Agency / constraint	`A3_mixed_agency`	Both agency-enhancing and agency-limiting elements
Agency / constraint	`A0_not_applicable`	Agency or constraint is not clearly discussed

Appendix C: Sentiment Analysis Method and Key Python Code

The sentiment analysis stage used platform-linked evaluative markers to create a baseline polarity dataset. Steam reviews were classified through the recommendation label. App Store and Google Play reviews were classified through star ratings. Ratings of 4-5 were treated as positive, rating 3 as neutral, and ratings of 1-2 as negative.

C.1 Rating-to-Sentiment Mapping

```
SENTIMENT_SCORE = {
    "positive": 1,
    "neutral": 0,
    "negative": -1,
    "mixed": 0,
    "unknown": 0,
}
```

```
def label_from_rating(rating):
    try:
        value = int(rating)
    except (TypeError, ValueError):
        return "unknown"
    if value >= 4:
        return "positive"
    if value <= 2:
        return "negative"
    return "neutral"
```

C.2 Platform Normalisation

```
def normalize_steam(rows, game):
    normalized = []
    for row in rows:
        label = row.get("sentiment_label") or "unknown"
        normalized.append({
            "game": game,
            "source": "steam",
            "language": row.get("language"),
            "record_id": row.get("recommendationid"),
            "text": row.get("review"),
            "rating": None,
            "sentiment_label": label,
            "sentiment_score": SENTIMENT_SCORE.get(label, 0),
            "label_source": "steam_recommendation",
        })
    return normalized

def normalize_app_store(rows):
    normalized = []
    for row in rows:
        label = label_from_rating(row.get("rating"))
        normalized.append({
            "game": "Love and Deepspace",
            "source": "app_store",
            "language": row.get("language"),
            "record_id": row.get("review_id"),
            "text": row.get("review"),
            "rating": row.get("rating"),
            "sentiment_label": label,
            "sentiment_score": SENTIMENT_SCORE.get(label, 0),
            "label_source": "app_store_rating",
        })
    return normalized

def normalize_google_play(rows):
    normalized = []
    for row in rows:
        label = label_from_rating(row.get("score"))
```

```
        normalized.append({
            "game": "Love and Deepspace",
            "source": "google_play",
            "language": row.get("language"),
            "record_id": row.get("review_id"),
            "text": row.get("review"),
            "rating": row.get("score"),
            "sentiment_label": label,
            "sentiment_score": SENTIMENT_SCORE.get(label, 0),
            "label_source": "google_play_rating",
        })
    return normalized
```

C.3 Summary Calculation

```
def build_summary(rows, group_fields):
    grouped = defaultdict(list)
    for row in rows:
        key = tuple(row.get(field) for field in group_fields)
        grouped[key].append(row)

    summary = []
    for key, group_rows in sorted(grouped.items()):
        counts = Counter(row["sentiment_label"] for row in group_rows)
        total = len(group_rows)
        avg_score = sum(row["sentiment_score"] for row in group_rows)
/ total if total else 0
        output = {field: value for field, value in zip(group_fields,
key)}
        output.update({
            "total": total,
            "positive": counts.get("positive", 0),
            "neutral": counts.get("neutral", 0),
            "negative": counts.get("negative", 0),
            "positive_pct": round(counts.get("positive", 0) / total *
100, 2) if total else 0,
            "neutral_pct": round(counts.get("neutral", 0) / total *
100, 2) if total else 0,
            "negative_pct": round(counts.get("negative", 0) / total *
100, 2) if total else 0,
            "avg_sentiment_score": round(avg_score, 4),
        })
        summary.append(output)
    return summary
```

C.4 Key Baseline Results

Game
Hakuoki: Kyoto Winds
Love and Deepspace
Monster Prom

Appendix D: Supplementary Lexical and Topic-Modelling Materials

This appendix records the supplementary word and topic materials used to support the main thematic analysis. No separate word-cloud figure is required for the dissertation because the topic-modelling summary table already presents the most useful lexical evidence in a more precise and readable form.

D.1 Aspect Keyword Examples

Aspect	Example keywords
Narrative	story, plot, route, ending, chapter, pacing, narrative, 剧情, 故事, 路线, 结局, ストーリー, ルート
Romance	romance, relationship, love interest, chemistry, intimacy, date, 恋爱, 男主, 陪伴, 恋愛, 彼氏, 推し
Gameplay	gameplay, mechanic, choice, combat, system, interaction, play, 玩法, 选择, 战斗, 系统, 選択肢, 戦闘

D.2 Dominant Topic Terms by Game and Language

Game	English	Chinese	Japanese
Hakuoki: Kyoto Winds	route, story, characters, romance, Edo, history	剧情, 历史, 新选组, 路线, 攻略, 结局	新選組, 主人公, キャラ, ルート, 攻略, 歴史
Love and Deepspace	story, characters, cards, storage, update, download	剧情, 男主, 抽卡, 资源, 活动, 更新	ストーリー, キャラクター, 声優, 戦闘, 課金, ダウンロード
Monster Prom	fun, friends, dating, characters, endings, prom	攻略, 路线, 选项, 结局, 数值, 有趣	Japanese subset too small for stable interpretation

These lexical patterns support the dissertation's cross-case interpretation. Hakuoki reviews cluster around routes, history, and story structure; Love and Deepspace reviews combine romance vocabulary with platform, update, storage, card, and resource terms; Monster Prom reviews foreground fun, choices, endings, strategy, and replay.

Appendix E: Representative Player Review Samples (Anonymised and Paraphrased)

The following examples are paraphrased from the coded review sample. Usernames, profile links, and other personal identifiers are not retained. The purpose of this appendix is to show how selected reviews informed the thematic interpretation without reproducing full user-generated texts.

Sample	Game	Source	Language	Sentiment	Paraphras	Analytical

					ed review content	relevance
E1	Hakuoki: Kyoto Winds	Steam	English	Positive	The reviewer describes the game as a major otome title with strong historical grounding, but also notes that the romance can feel less immediate than expected.	Historical worldbuilding supports attachment while route pacing can produce frustration.
E2	Hakuoki: Kyoto Winds	Steam	English	Positive	The reviewer recommends the game but says the first part feels incomplete without the companion title and that repeated play is needed to access the fuller story.	Route-based slow-burn romance creates both narrative investment and delayed payoff.
E3	Hakuoki: Kyoto Winds	Steam	English	Negative	The reviewer finds the historical	Demonstrates how historical density

					tragedy emotionally exhausting and feels that the expected romance is overwhelmed by a long, depressing story arc.	can become emotional strain.
E4	Love and Deepspace	App Store	English	Positive	The reviewer praises the high production quality, female-oriented address, character attachment, customisable protagonist, card stories, and dates, while also mentioning storage and gacha constraints.	Shows attachment and intimacy operating alongside mobile-system limits.
E5	Love and Deepspace	App Store	English	Neutral	The reviewer argues that the game is not friendly to free-to-play users, criticising	Illustrates frustration with monetised and live-service access to romance.

					banner pressure, FOMO, rerun structure, and limited resources.	
E6	Love and Deepspace	App Store	English	Negative	The reviewer returns because of attachment to a specific love interest, but criticises daily grind, spending pressure, and lore being locked behind paywalled or limited content.	Shows strong character attachment under conditions of restricted access.
E7	Monster Prom	Steam	English	Positive	The reviewer praises the game's event variety, stat-linked choices, multiplayer play, endings, and art style, while noting that random event	Demonstrates mixed agency in procedural romance.

					selection and save limitations can affect agency.	
E8	Monster Prom	Steam	English	Negative	The reviewer likes the style and characters but feels that opaque stat requirements make romantic success guide-dependent and repetitive.	Shows how procedural uncertainty can become frustration.
E9	Monster Prom	Steam	English	Positive	The reviewer describes the game as funny and replayable, with pleasure coming from silly experimentation and repeated pursuit of different outcomes.	Supports the interpretation of romance as replay-based procedural possibility.